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## Course: CME 435.3 Verification of Digital Systems

## Lab #: 4: UVM Testbench Implementation

Instructions to Run Provide step-by-step instructions to run each part of the lab project:

- Part 1 (Compilation): Navigate to the script directory and execute run.csh to compile the testbench framework components (test, environment, agents).
    - csh script/run.csh

- Part 2 (Sanity Test): Execute the run.csh script specifying pdm_sanity_test to run the directed sanity test in either batch mode or GUI mode.

- Part 3 (Additional Tests): Execute the run.csh script to run the additional constrained random tests. The specific tests implemented include pdm_packet_len_test, pdm_proceed_delay_test, pdm_saturation_test, pdm_reset_test, and pdm_random_test. Use the appropriate flags in the script to run these in GUI mode to dump and view waves.

Questions & Answers Provide answers to any applicable questions:

1. Question 1: No specific questions were assigned in the CME435_lab_4_v1.pdf lab manual for this submission.

2. Bugs Identified: None

Results Provide results for each part below, making sure to include relevant screenshots:

# Running the tests

```
csh script/run.csh -b
csh script/run.csh -b -t pdm_sanity_test
csh script/run.csh -b -t pdm_packet_len_test
csh script/run.csh -b -t pdm_proceed_delay_test
csh script/run.csh -b -t pdm_saturation_test
csh script/run.csh -b -t pdm_reset_test
csh script/run.csh -b -t pdm_random_test
```

## Running pdm_test_base.sv terminal output

None

```
Directory 'work' already exists. Skipping vlib.
QuestaSim-64 vmap 2022.1_2 Lib Mapping Utility 2022.04 Apr  2 2022
vmap work work
Modifying /home/.engr-ece/rfp479/Downloads/01_CME_435_L4/lab_submission/modelsim.ini
Compiling design and verification files...
QuestaSim-64 vlog 2022.1_2 Compiler 2022.04 Apr  2 2022
Start time: 15:19:47 on Nov 08,2026
vlog -sv -L mtiUvm
"+incdir+/home/.engr-ece/rfp479/Downloads/01_CME_435_L4/lab_submission/verification"
/home/.engr-ece/rfp479/Downloads/01_CME_435_L4/lab_submission/dut/pdm_core.svp
/home/.engr-ece/rfp479/Downloads/01_CME_435_L4/lab_submission/dut/pdm_top.v
/home/.engr-ece/rfp479/Downloads/01_CME_435_L4/lab_submission/verification/pdm_upstream_if.sv
/home/.engr-ece/rfp479/Downloads/01_CME_435_L4/lab_submission/verification/pdm_downstream_if.sv
/home/.engr-ece/rfp479/Downloads/01_CME_435_L4/lab_submission/verification/pdm_pkg.sv
/home/.engr-ece/rfp479/Downloads/01_CME_435_L4/lab_submission/verification/top.sv
-- Compiling module pdm
-- Compiling interface pdm_upstream_if
-- Compiling interface pdm_downstream_if
-- Compiling package pdm_pkg
-- Importing package mtiUvm.uvm_pkg (uvm-1.1d Built-in)
** Note: (vlog-2286)
/home/.engr-ece/rfp479/Downloads/01_CME_435_L4/lab_submission/verification/pdm_pkg.sv(8): Using
implicit
+incdir+/CMC/tools/siemens/questasim_2022.1_2/questasim/uvm-1.1d/../verilog_src/uvm-1.1d/src
from import uvm_pkg
-- Compiling module top
-- Importing package pdm_pkg

Top level modules:
    top
End time: 15:19:47 on Nov 08,2026, Elapsed time: 0:00:00
Errors: 0, Warnings: 0
Starting Questa Simulator in BATCH mode for test: pdm_test_base...
Reading pref.tcl

# 2022.1_2

# set TESTNAME pdm_test_base
# pdm_test_base
#  do /home/.engr-ece/rfp479/Downloads/01_CME_435_L4/lab_submission/script/run.do
# ============================================================
#   RUN.DO: Launching UVM test: pdm_test_base
# ============================================================
# vsim -L mtiUvm work.top "+UVM_TESTNAME=pdm_test_base" "+UVM_NO_RELNOTES"
# Start time: 15:19:48 on Nov 08,2026
# ** Note: (vsim-8009) Loading existing optimized design _opt
# //  Questa Sim-64
# //  Version 2022.1_2 linux_x86_64 Apr  2 2022
```

```
# //
# //  Copyright 1991-2022 Mentor Graphics Corporation
# //  All Rights Reserved.
# //
# //  QuestaSim and its associated documentation contain trade
# //  secrets and commercial or financial information that are the property of
# //  Mentor Graphics Corporation and are privileged, confidential,
# //  and exempt from disclosure under the Freedom of Information Act,
# //  5 U.S.C. Section 552. Furthermore, this information
# //  is prohibited from disclosure under the Trade Secrets Act,
# //  18 U.S.C. Section 1905.
# //
# Loading sv_std.std
# Loading work.pdm_upstream_if(fast)
# Loading work.pdm_downstream_if(fast)
# Loading mtiUvm.uvm_pkg(fast)
# Loading work.pdm_pkg(fast)
# Loading mtiUvm.questa_uvm_pkg(fast)
# Loading work.top(fast)
# Loading work.pdm_downstream_if(fast__2)
# Loading /CMC/tools/siemens/questasim_2022.1_2/questasim/uvm-1.1d/linux_x86_64/uvm_dpi.so
# ----------------------------------------------------------------
# UVM-1.1d
# (C) 2007-2013 Mentor Graphics Corporation
# (C) 2007-2013 Cadence Design Systems, Inc.
# (C) 2006-2013 Synopsys, Inc.
# (C) 2011-2013 Cypress Semiconductor Corp.
# ----------------------------------------------------------------
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(277) @ 0: reporter [Questa UVM]
QUESTA_UVM-1.2.3
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(278) @ 0: reporter [Questa UVM]
questa_uvm::init(+struct)
# UVM_INFO @ 0: reporter [RNTST] Running test pdm_test_base...
# UVM_INFO @ 0: reporter [UVMTOP] UVM testbench topology:
# ----------------------------------------------------------------
# Name                      Type                        Size  Value
# ----------------------------------------------------------------
# uvm_test_top              pdm_test_base               -     @485
#   env                     pdm_env                     -     @507
#     downstream_agent_0    pdm_downstream_agent        -     @529
#       driver              pdm_downstream_driver       -     @743
#         rsp_port          uvm_analysis_port           -     @758
#         seq_item_port     uvm_seq_item_pull_port      -     @750
#       monitor             pdm_downstream_monitor      -     @564
#         downstream_txn_ap uvm_analysis_port           -     @579
#         req_ap            uvm_analysis_port           -     @571
#       sequencer           pdm_downstream_sequencer    -     @587
#         mon_req_fifo      uvm_tlm_analysis_fifo #(T)  -     @696
#           analysis_export uvm_analysis_imp            -     @735
#           get_ap          uvm_analysis_port           -     @727
```

```
#           get_peek_export  uvm_get_peek_imp           -     @711
#           put_ap           uvm_analysis_port          -     @719
#           put_export       uvm_put_imp                -     @703
#         rsp_export         uvm_analysis_export        -     @594
#         seq_item_export    uvm_seq_item_pull_imp      -     @688
#         arbitration_queue  array                      0     -
#         lock_queue         array                      0     -
#         num_last_reqs      integral                   32    'd1
#         num_last_rsps      integral                   32    'd1
#     downstream_agent_1     pdm_downstream_agent       -     @537
#       driver               pdm_downstream_driver      -     @956
#         rsp_port           uvm_analysis_port          -     @971
#         seq_item_port      uvm_seq_item_pull_port     -     @963
#       monitor              pdm_downstream_monitor     -     @777
#         downstream_txn_ap  uvm_analysis_port          -     @792
#         req_ap             uvm_analysis_port          -     @784
#       sequencer            pdm_downstream_sequencer   -     @800
#         mon_req_fifo       uvm_tlm_analysis_fifo #(T) -     @909
#           analysis_export  uvm_analysis_imp           -     @948
#           get_ap           uvm_analysis_port          -     @940
#           get_peek_export  uvm_get_peek_imp           -     @924
#           put_ap           uvm_analysis_port          -     @932
#           put_export       uvm_put_imp                -     @916
#         rsp_export         uvm_analysis_export        -     @807
#         seq_item_export    uvm_seq_item_pull_imp      -     @901
#         arbitration_queue  array                      0     -
#         lock_queue         array                      0     -
#         num_last_reqs      integral                   32    'd1
#         num_last_rsps      integral                   32    'd1
#     downstream_agent_2     pdm_downstream_agent       -     @545
#       driver               pdm_downstream_driver      -     @1169
#         rsp_port           uvm_analysis_port          -     @1184
#         seq_item_port      uvm_seq_item_pull_port     -     @1176
#       monitor              pdm_downstream_monitor     -     @990
#         downstream_txn_ap  uvm_analysis_port          -     @1005
#         req_ap             uvm_analysis_port          -     @997
#       sequencer            pdm_downstream_sequencer   -     @1013
#         mon_req_fifo       uvm_tlm_analysis_fifo #(T) -     @1122
#           analysis_export  uvm_analysis_imp           -     @1161
#           get_ap           uvm_analysis_port          -     @1153
#           get_peek_export  uvm_get_peek_imp           -     @1137
#           put_ap           uvm_analysis_port          -     @1145
#           put_export       uvm_put_imp                -     @1129
#         rsp_export         uvm_analysis_export        -     @1020
#         seq_item_export    uvm_seq_item_pull_imp      -     @1114
#         arbitration_queue  array                      0     -
#         lock_queue         array                      0     -
#         num_last_reqs      integral                   32    'd1
#         num_last_rsps      integral                   32    'd1
#     downstream_agent_3     pdm_downstream_agent       -     @553
```

```
#       driver               pdm_downstream_driver       -     @1382
#         rsp_port           uvm_analysis_port           -     @1397
#         seq_item_port      uvm_seq_item_pull_port      -     @1389
#       monitor              pdm_downstream_monitor      -     @1203
#         downstream_txn_ap  uvm_analysis_port           -     @1218
#         req_ap             uvm_analysis_port           -     @1210
#       sequencer            pdm_downstream_sequencer    -     @1226
#         mon_req_fifo       uvm_tlm_analysis_fifo #(T)  -     @1335
#           analysis_export  uvm_analysis_imp            -     @1374
#           get_ap           uvm_analysis_port           -     @1366
#           get_peek_export  uvm_get_peek_imp            -     @1350
#           put_ap           uvm_analysis_port           -     @1358
#           put_export       uvm_put_imp                 -     @1342
#         rsp_export         uvm_analysis_export         -     @1233
#         seq_item_export    uvm_seq_item_pull_imp       -     @1327
#         arbitration_queue  array                       0     -
#         lock_queue         array                       0     -
#         num_last_reqs      integral                    32    'd1
#         num_last_rsps      integral                    32    'd1
#     upstream_agent         pdm_upstream_agent          -     @519
#       driver               pdm_upstream_driver         -     @1540
#         rsp_port           uvm_analysis_port           -     @1555
#         seq_item_port      uvm_seq_item_pull_port      -     @1547
#       monitor              pdm_upstream_monitor        -     @1416
#         upstream_txn_ap    uvm_analysis_port           -     @1423
#       sequencer            uvm_sequencer               -     @1431
#         rsp_export         uvm_analysis_export         -     @1438
#         seq_item_export    uvm_seq_item_pull_imp       -     @1532
#         arbitration_queue  array                       0     -
#         lock_queue         array                       0     -
#         num_last_reqs      integral                    32    'd1
#         num_last_rsps      integral                    32    'd1
# ----------------------------------------------------------------
#
# UVM_INFO verilog_src/uvm-1.1d/src/base/uvm_objection.svh(1267) @ 200000: reporter [TEST_DONE]
'run' phase is ready to proceed to the 'extract' phase
#
# ========================================================
#                  TEST EXECUTION SUMMARY
# ========================================================
#   TEST NAME    : pdm_test_base
#   SIM TIME     : 200000
#   FATAL COUNT  : 0
#   ERROR COUNT  : 0
#   WARNING COUNT: 0
#   INFO COUNT   : 5
# --------------------------------------------------------
#               *** UVM TEST PASSED ***
# ========================================================
#
```

```
#
# --- UVM Report Summary ---
#
# ** Report counts by severity
# UVM_INFO :    5
# UVM_WARNING :    0
# UVM_ERROR :    0
# UVM_FATAL :    0
# ** Report counts by id
# [Questa UVM]     2
# [RNTST]     1
# [TEST_DONE]     1
# [UVMTOP]     1
# ** Note: $finish    :
/CMC/tools/siemens/questasim_2022.1_2/questasim/linux_x86_64/../verilog_src/uvm-1.1d/src/base/uvm_root.svh(430)
#    Time: 200 ns  Iteration: 54  Instance: /top
# End time: 15:19:49 on Nov 08,2026, Elapsed time: 0:00:01
# Errors: 0, Warnings: 0
```

## Running pdm_sanity_test.sv terminal output

None

```
Directory 'work' already exists. Skipping vlib.
QuestaSim-64 vmap 2022.1_2 Lib Mapping Utility 2022.04 Apr  2 2022
vmap work work
Modifying
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/modelsim.ini
Compiling design and verification files...
QuestaSim-64 vlog 2022.1_2 Compiler 2022.04 Apr  2 2022
Start time: 14:59:33 on Nov 08,2026
vlog -sv -L mtiUvm
"+incdir+/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification"
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/dut/pdm_core.svp
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/dut/pdm_top.v
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_upstream_if.sv
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_downstream_if.sv
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_pkg.sv
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/top.sv
-- Compiling module pdm
```

```
-- Compiling interface pdm_upstream_if
-- Compiling interface pdm_downstream_if
-- Compiling package pdm_pkg
-- Importing package mtiUvm.uvm_pkg (uvm-1.1d Built-in)
** Note: (vlog-2286)
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
pkg.sv(8): Using implicit
+incdir+/CMC/tools/siemens/questasim_2022.1_2/questasim/uvm-1.1d/../verilog_src/uvm-1.1d/src
from import uvm_pkg
-- Compiling module top
-- Importing package pdm_pkg

Top level modules:
        top
End time: 14:59:34 on Nov 08,2026, Elapsed time: 0:00:01
Errors: 0, Warnings: 0
Starting Questa Simulator in BATCH mode for test: pdm_sanity_test...
Reading pref.tcl

# 2022.1_2

# set TESTNAME pdm_sanity_test
# pdm_sanity_test
#  do
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/script/run.do
# ============================================================
#   RUN.DO: Launching UVM test: pdm_sanity_test
# ============================================================
# vsim -L mtiUvm work.top "+UVM_TESTNAME=pdm_sanity_test" "+UVM_NO_RELNOTES"
# Start time: 14:59:35 on Nov 08,2026
# ** Note: (vsim-8009) Loading existing optimized design _opt2
# //  Questa Sim-64
# //  Version 2022.1_2 linux_x86_64 Apr  2 2022
# //
# //  Copyright 1991-2022 Mentor Graphics Corporation
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# //
# //  QuestaSim and its associated documentation contain trade
# //  secrets and commercial or financial information that are the property of
# //  Mentor Graphics Corporation and are privileged, confidential,
# //  and exempt from disclosure under the Freedom of Information Act,
# //  5 U.S.C. Section 552. Furthermore, this information
# //  is prohibited from disclosure under the Trade Secrets Act,
# //  18 U.S.C. Section 1905.
# //
# Loading sv_std.std
# Loading work.pdm_upstream_if(fast)
# Loading work.pdm_downstream_if(fast)
# Loading mtiUvm.uvm_pkg(fast)
# Loading work.pdm_pkg(fast)
```

```
# Loading mtiUvm.questa_uvm_pkg(fast)
# Loading work.top(fast)
# Loading work.pdm_downstream_if(fast__2)
# Loading /CMC/tools/siemens/questasim_2022.1_2/questasim/uvm-1.1d/linux_x86_64/uvm_dpi.so
# ----------------------------------------------------------------
# UVM-1.1d
# (C) 2007-2013 Mentor Graphics Corporation
# (C) 2007-2013 Cadence Design Systems, Inc.
# (C) 2006-2013 Synopsys, Inc.
# (C) 2011-2013 Cypress Semiconductor Corp.
# ----------------------------------------------------------------
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(277) @ 0: reporter [Questa UVM]
QUESTA_UVM-1.2.3
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(278) @ 0: reporter [Questa UVM]
questa_uvm::init(+struct)
# UVM_INFO @ 0: reporter [RNTST] Running test pdm_sanity_test...
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
sanity_test.sv(21) @ 0: uvm_test_top [pdm_sanity_test] Waiting for reset deassertion (rst_b ==
1)...
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
sanity_test.sv(28) @ 110000: uvm_test_top [pdm_sanity_test] Executing pdm_sanity_seq on upstream
sequencer...
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
sanity_seq.sv(14) @ 110000: uvm_test_top.env.upstream_agent.sequencer@@seq [pdm_sanity_seq]
Starting directed sanity sequence across Ports 1 to 4
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
sanity_seq.sv(33) @ 120000: uvm_test_top.env.upstream_agent.sequencer@@seq [pdm_sanity_seq]
Sending sanity packet to Port 1 (Length: 4 bytes)
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
sanity_seq.sv(33) @ 280000: uvm_test_top.env.upstream_agent.sequencer@@seq [pdm_sanity_seq]
Sending sanity packet to Port 2 (Length: 8 bytes)
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
sanity_seq.sv(33) @ 520000: uvm_test_top.env.upstream_agent.sequencer@@seq [pdm_sanity_seq]
Sending sanity packet to Port 3 (Length: 12 bytes)
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
sanity_seq.sv(33) @ 840000: uvm_test_top.env.upstream_agent.sequencer@@seq [pdm_sanity_seq]
Sending sanity packet to Port 4 (Length: 16 bytes)
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
sanity_seq.sv(39) @ 1220000: uvm_test_top.env.upstream_agent.sequencer@@seq [pdm_sanity_seq]
Sanity sequence completed driving all 4 packets
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
```

```
sanity_test.sv(33) @ 1220000: uvm_test_top [pdm_sanity_test] Waiting for final downstream
handshakes to drain...
# UVM_INFO verilog_src/uvm-1.1d/src/base/uvm_objection.svh(1267) @ 1720000: reporter [TEST_DONE]
'run' phase is ready to proceed to the 'extract' phase
#
# ------------------------------------------------------
#                  TEST EXECUTION SUMMARY
# ------------------------------------------------------
#   TEST NAME    : pdm_sanity_test
#   SIM TIME     : 1720000
#   FATAL COUNT  : 0
#   ERROR COUNT  : 0
#   WARNING COUNT: 0
#   INFO COUNT   : 13
# ------------------------------------------------------
#               *** UVM TEST PASSED ***
# ------------------------------------------------------
#
#
# --- UVM Report Summary ---
#
# ** Report counts by severity
# UVM_INFO :   13
# UVM_WARNING :    0
# UVM_ERROR :    0
# UVM_FATAL :    0
# ** Report counts by id
# [Questa UVM]     2
# [RNTST]     1
# [TEST_DONE]     1
# [pdm_sanity_seq]     6
# [pdm_sanity_test]     3
# ** Note: $finish    :
/CMC/tools/siemens/questasim_2022.1_2/questasim/linux_x86_64/../verilog_src/uvm-1.1d/src/base/uv
m_root.svh(430)
#    Time: 1720 ns  Iteration: 54  Instance: /top
# End time: 14:59:36 on Nov 08,2026, Elapsed time: 0:00:01
# Errors: 0, Warnings: 0
```

## Running pdm_packet_len_test.sv terminal output

None

```
Directory 'work' already exists. Skipping vlib.
QuestaSim-64 vmap 2022.1_2 Lib Mapping Utility 2022.04 Apr  2 2022
vmap work work
Modifying
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/modelsim.ini
```

```
Compiling design and verification files...
QuestaSim-64 vlog 2022.1_2 Compiler 2022.04 Apr  2 2022
Start time: 15:01:32 on Nov 08,2026
vlog -sv -L mtiUvm
"+incdir+/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verifica
tion"
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/dut/pdm_core.svp
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/dut/pdm_top.v
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
upstream_if.sv
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
downstream_if.sv
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
pkg.sv
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/top.
sv
-- Compiling module pdm
-- Compiling interface pdm_upstream_if
-- Compiling interface pdm_downstream_if
-- Compiling package pdm_pkg
-- Importing package mtiUvm.uvm_pkg (uvm-1.1d Built-in)
** Note: (vlog-2286)
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
pkg.sv(8): Using implicit
+incdir+/CMC/tools/siemens/questasim_2022.1_2/questasim/uvm-1.1d/../verilog_src/uvm-1.1d/src
from import uvm_pkg
-- Compiling module top
-- Importing package pdm_pkg

Top level modules:
        top
End time: 15:01:32 on Nov 08,2026, Elapsed time: 0:00:00
Errors: 0, Warnings: 0
Starting Questa Simulator in BATCH mode for test: pdm_packet_len_test...
Reading pref.tcl

# 2022.1_2

# set TESTNAME pdm_packet_len_test
# pdm_packet_len_test
#  do
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/script/run.do
# ============================================================
#   RUN.DO: Launching UVM test: pdm_packet_len_test
# ============================================================
# vsim -L mtiUvm work.top "+UVM_TESTNAME=pdm_packet_len_test" "+UVM_NO_RELNOTES"
# Start time: 15:01:33 on Nov 08,2026
# ** Note: (vsim-8009) Loading existing optimized design _opt2
# //  Questa Sim-64
# //  Version 2022.1_2 linux_x86_64 Apr  2 2022
```

```
# //
# //  Copyright 1991-2022 Mentor Graphics Corporation
# //  All Rights Reserved.
# //
# //  QuestaSim and its associated documentation contain trade
# //  secrets and commercial or financial information that are the property of
# //  Mentor Graphics Corporation and are privileged, confidential,
# //  and exempt from disclosure under the Freedom of Information Act,
# //  5 U.S.C. Section 552. Furthermore, this information
# //  is prohibited from disclosure under the Trade Secrets Act,
# //  18 U.S.C. Section 1905.
# //
# Loading sv_std.std
# Loading work.pdm_upstream_if(fast)
# Loading work.pdm_downstream_if(fast)
# Loading mtiUvm.uvm_pkg(fast)
# Loading work.pdm_pkg(fast)
# Loading mtiUvm.questa_uvm_pkg(fast)
# Loading work.top(fast)
# Loading work.pdm_downstream_if(fast__2)
# Loading /CMC/tools/siemens/questasim_2022.1_2/questasim/uvm-1.1d/linux_x86_64/uvm_dpi.so
# ----------------------------------------------------------------
# UVM-1.1d
# (C) 2007-2013 Mentor Graphics Corporation
# (C) 2007-2013 Cadence Design Systems, Inc.
# (C) 2006-2013 Synopsys, Inc.
# (C) 2011-2013 Cypress Semiconductor Corp.
# ----------------------------------------------------------------
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(277) @ 0: reporter [Questa UVM]
QUESTA_UVM-1.2.3
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(278) @ 0: reporter [Questa UVM]
questa_uvm::init(+struct)
# UVM_INFO @ 0: reporter [RNTST] Running test pdm_packet_len_test...
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
packet_len_test.sv(25) @ 110000: uvm_test_top [pdm_packet_len_test] --- Phase 1: Directed length
sweep across all ports ---
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
pkt_len_seq.sv(27) @ 110000: uvm_test_top.env.upstream_agent.sequencer@@len_seq
[pdm_pkt_len_seq] Starting packet length test sequence in mode SWEEP_ALL
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
packet_len_test.sv(31) @ 4580000: uvm_test_top [pdm_packet_len_test] --- Phase 2: Min (4B) and
Max (16B) corner-case length testing ---
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
pkt_len_seq.sv(27) @ 4580000: uvm_test_top.env.upstream_agent.sequencer@@len_seq
[pdm_pkt_len_seq] Starting packet length test sequence in mode CORNER_MIN_MAX
```

```
# UVM_INFO verilog_src/uvm-1.1d/src/base/uvm_objection.svh(1267) @ 8680000: reporter [TEST_DONE]
'run' phase is ready to proceed to the 'extract' phase
#
# ------------------------------------------------------
#                  TEST EXECUTION SUMMARY
# ------------------------------------------------------
#   TEST NAME    : pdm_packet_len_test
#   SIM TIME     : 8680000
#   FATAL COUNT  : 0
#   ERROR COUNT  : 0
#   WARNING COUNT: 0
#   INFO COUNT   : 8
# ------------------------------------------------------
#               *** UVM TEST PASSED ***
# ------------------------------------------------------
#
#
# --- UVM Report Summary ---
#
# ** Report counts by severity
# UVM_INFO :    8
# UVM_WARNING :    0
# UVM_ERROR :    0
# UVM_FATAL :    0
# ** Report counts by id
# [Questa UVM]     2
# [RNTST]     1
# [TEST_DONE]     1
# [pdm_packet_len_test]     2
# [pdm_pkt_len_seq]     2
# ** Note: $finish    :
/CMC/tools/siemens/questasim_2022.1_2/questasim/linux_x86_64/../verilog_src/uvm-1.1d/src/base/uv
m_root.svh(430)
#    Time: 8680 ns  Iteration: 54  Instance: /top
# End time: 15:01:35 on Nov 08,2026, Elapsed time: 0:00:02
# Errors: 0, Warnings: 0
```

## Running pdm_proceed_delay_test.sv terminal output

None

```
Directory 'work' already exists. Skipping vlib.
QuestaSim-64 vmap 2022.1_2 Lib Mapping Utility 2022.04 Apr  2 2022
vmap work work
Modifying
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/modelsim.ini
Compiling design and verification files...
QuestaSim-64 vlog 2022.1_2 Compiler 2022.04 Apr  2 2022
```

```
Start time: 15:02:55 on Nov 08,2026
vlog -sv -L mtiUvm
"+incdir+/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verifica
tion"
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/dut/pdm_core.svp
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/dut/pdm_top.v
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
upstream_if.sv
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
downstream_if.sv
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
pkg.sv
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/top.
sv
-- Compiling module pdm
-- Compiling interface pdm_upstream_if
-- Compiling interface pdm_downstream_if
-- Compiling package pdm_pkg
-- Importing package mtiUvm.uvm_pkg (uvm-1.1d Built-in)
** Note: (vlog-2286)
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
pkg.sv(8): Using implicit
+incdir+/CMC/tools/siemens/questasim_2022.1_2/questasim/uvm-1.1d/../verilog_src/uvm-1.1d/src
from import uvm_pkg
-- Compiling module top
-- Importing package pdm_pkg

Top level modules:
        top
End time: 15:02:55 on Nov 08,2026, Elapsed time: 0:00:00
Errors: 0, Warnings: 0
Starting Questa Simulator in BATCH mode for test: pdm_proceed_delay_test...
Reading pref.tcl

# 2022.1_2

# set TESTNAME pdm_proceed_delay_test
# pdm_proceed_delay_test
#  do
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/script/run.do
# ============================================================
#   RUN.DO: Launching UVM test: pdm_proceed_delay_test
# ============================================================
# vsim -L mtiUvm work.top "+UVM_TESTNAME=pdm_proceed_delay_test" "+UVM_NO_RELNOTES"
# Start time: 15:02:56 on Nov 08,2026
# ** Note: (vsim-8009) Loading existing optimized design _opt2
# //  Questa Sim-64
# //  Version 2022.1_2 linux_x86_64 Apr  2 2022
# //
# //  Copyright 1991-2022 Mentor Graphics Corporation
```

```
# //  All Rights Reserved.
# //
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# //  and exempt from disclosure under the Freedom of Information Act,
# //  5 U.S.C. Section 552. Furthermore, this information
# //  is prohibited from disclosure under the Trade Secrets Act,
# //  18 U.S.C. Section 1905.
# //
# Loading sv_std.std
# Loading work.pdm_upstream_if(fast)
# Loading work.pdm_downstream_if(fast)
# Loading mtiUvm.uvm_pkg(fast)
# Loading work.pdm_pkg(fast)
# Loading mtiUvm.questa_uvm_pkg(fast)
# Loading work.top(fast)
# Loading work.pdm_downstream_if(fast__2)
# Loading /CMC/tools/siemens/questasim_2022.1_2/questasim/uvm-1.1d/linux_x86_64/uvm_dpi.so
# ----------------------------------------------------------------
# UVM-1.1d
# (C) 2007-2013 Mentor Graphics Corporation
# (C) 2007-2013 Cadence Design Systems, Inc.
# (C) 2006-2013 Synopsys, Inc.
# (C) 2011-2013 Cypress Semiconductor Corp.
# ----------------------------------------------------------------
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(277) @ 0: reporter [Questa UVM]
QUESTA_UVM-1.2.3
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(278) @ 0: reporter [Questa UVM]
questa_uvm::init(+struct)
# UVM_INFO @ 0: reporter [RNTST] Running test pdm_proceed_delay_test...
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
proceed_delay_test.sv(24) @ 110000: uvm_test_top [pdm_proceed_delay_test] --- Phase 1: Testing
minimum PROCEED delay (1 clock cycle) ---
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
random_traffic_seq.sv(20) @ 110000: uvm_test_top.env.upstream_agent.sequencer@@traffic_seq
[pdm_random_traffic_seq] Generating 8 constrained-random packets
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
proceed_delay_test.sv(35) @ 2380000: uvm_test_top [pdm_proceed_delay_test] --- Phase 2: Testing
maximum PROCEED delay (4 clock cycles) ---
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
random_traffic_seq.sv(20) @ 2380000: uvm_test_top.env.upstream_agent.sequencer@@traffic_seq
[pdm_random_traffic_seq] Generating 8 constrained-random packets
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
```

```
proceed_delay_test.sv(46) @ 4980000: uvm_test_top [pdm_proceed_delay_test] --- Phase 3: Testing
randomized PROCEED delays (1 to 4 cycles) ---
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
random_traffic_seq.sv(20) @ 4980000: uvm_test_top.env.upstream_agent.sequencer@@traffic_seq
[pdm_random_traffic_seq] Generating 12 constrained-random packets
# UVM_INFO verilog_src/uvm-1.1d/src/base/uvm_objection.svh(1267) @ 8580000: reporter [TEST_DONE]
'run' phase is ready to proceed to the 'extract' phase
#
# ============================================================
#                  TEST EXECUTION SUMMARY
# ============================================================
#   TEST NAME    : pdm_proceed_delay_test
#   SIM TIME     : 8580000
#   FATAL COUNT  : 0
#   ERROR COUNT  : 0
#   WARNING COUNT: 0
#   INFO COUNT   : 10
# ------------------------------------------------------------
#               *** UVM TEST PASSED ***
# ============================================================
#
#
# --- UVM Report Summary ---
#
# ** Report counts by severity
# UVM_INFO :   10
# UVM_WARNING :    0
# UVM_ERROR :    0
# UVM_FATAL :    0
# ** Report counts by id
# [Questa UVM]     2
# [RNTST]     1
# [TEST_DONE]     1
# [pdm_proceed_delay_test]     3
# [pdm_random_traffic_seq]     3
# ** Note: $finish    :
/CMC/tools/siemens/questasim_2022.1_2/questasim/linux_x86_64/../verilog_src/uvm-1.1d/src/base/uv
m_root.svh(430)
#    Time: 8580 ns  Iteration: 54  Instance: /top
# End time: 15:02:58 on Nov 08,2026, Elapsed time: 0:00:02
# Errors: 0, Warnings: 0
```

Running pdm_saturation_test.sv terminal output

None

```
Directory 'work' already exists. Skipping vlib.
QuestaSim-64 vmap 2022.1_2 Lib Mapping Utility 2022.04 Apr  2 2022
vmap work work
Modifying
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/modelsim.ini
Compiling design and verification files...
QuestaSim-64 vlog 2022.1_2 Compiler 2022.04 Apr  2 2022
Start time: 15:03:41 on Nov 08,2026
vlog -sv -L mtiUvm
"+incdir+/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verifica
tion"
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/dut/pdm_core.svp
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/dut/pdm_top.v
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
upstream_if.sv
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
downstream_if.sv
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
pkg.sv
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/top.
sv
-- Compiling module pdm
-- Compiling interface pdm_upstream_if
-- Compiling interface pdm_downstream_if
-- Compiling package pdm_pkg
-- Importing package mtiUvm.uvm_pkg (uvm-1.1d Built-in)
** Note: (vlog-2286)
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
pkg.sv(8): Using implicit
+incdir+/CMC/tools/siemens/questasim_2022.1_2/questasim/uvm-1.1d/../verilog_src/uvm-1.1d/src
from import uvm_pkg
-- Compiling module top
-- Importing package pdm_pkg

Top level modules:
        top
End time: 15:03:41 on Nov 08,2026, Elapsed time: 0:00:00
Errors: 0, Warnings: 0
Starting Questa Simulator in BATCH mode for test: pdm_saturation_test...
Reading pref.tcl

# 2022.1_2

# set TESTNAME pdm_saturation_test
# pdm_saturation_test
#  do
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/script/run.do
# ============================================================
#   RUN.DO: Launching UVM test: pdm_saturation_test
# ============================================================
```

```
# vsim -L mtiUvm work.top "+UVM_TESTNAME=pdm_saturation_test" "+UVM_NO_RELNOTES"
# Start time: 15:03:42 on Nov 08,2026
# ** Note: (vsim-8009) Loading existing optimized design _opt2
# //  Questa Sim-64
# //  Version 2022.1_2 linux_x86_64 Apr  2 2022
# //
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# //
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# //  and exempt from disclosure under the Freedom of Information Act,
# //  5 U.S.C. Section 552. Furthermore, this information
# //  is prohibited from disclosure under the Trade Secrets Act,
# //  18 U.S.C. Section 1905.
# //
# Loading sv_std.std
# Loading work.pdm_upstream_if(fast)
# Loading work.pdm_downstream_if(fast)
# Loading mtiUvm.uvm_pkg(fast)
# Loading work.pdm_pkg(fast)
# Loading mtiUvm.questa_uvm_pkg(fast)
# Loading work.top(fast)
# Loading work.pdm_downstream_if(fast__2)
# Loading /CMC/tools/siemens/questasim_2022.1_2/questasim/uvm-1.1d/linux_x86_64/uvm_dpi.so
# ----------------------------------------------------------------
# UVM-1.1d
# (C) 2007-2013 Mentor Graphics Corporation
# (C) 2007-2013 Cadence Design Systems, Inc.
# (C) 2006-2013 Synopsys, Inc.
# (C) 2011-2013 Cypress Semiconductor Corp.
# ----------------------------------------------------------------
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(277) @ 0: reporter [Questa UVM]
QUESTA_UVM-1.2.3
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(278) @ 0: reporter [Questa UVM]
questa_uvm::init(+struct)
# UVM_INFO @ 0: reporter [RNTST] Running test pdm_saturation_test...
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
saturation_test.sv(22) @ 110000: uvm_test_top [pdm_saturation_test] Setting downstream PROCEED
delay to 4 cycles to induce pipeline backpressure
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
saturation_test.sv(32) @ 110000: uvm_test_top [pdm_saturation_test] --- Injecting saturating
burst of 32 packets ---
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
saturation_seq.sv(20) @ 110000: uvm_test_top.env.upstream_agent.sequencer@@sat_seq
[pdm_saturation_seq] Launching saturation burst of 32 back-to-back packets
```

```
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
saturation_seq.sv(38) @ 9540000: uvm_test_top.env.upstream_agent.sequencer@@sat_seq
[pdm_saturation_seq] Completed saturation burst generation
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
saturation_test.sv(36) @ 9540000: uvm_test_top [pdm_saturation_test] Saturation stimulus
complete. Draining downstream FIFOs...
# UVM_INFO verilog_src/uvm-1.1d/src/base/uvm_objection.svh(1267) @ 11040000: reporter
[TEST_DONE] 'run' phase is ready to proceed to the 'extract' phase
#
# ========================================================
#                  TEST EXECUTION SUMMARY
# ========================================================
#   TEST NAME    : pdm_saturation_test
#   SIM TIME     : 11040000
#   FATAL COUNT  : 0
#   ERROR COUNT  : 0
#   WARNING COUNT: 0
#   INFO COUNT   : 9
# --------------------------------------------------------
#               *** UVM TEST PASSED ***
# ========================================================
#
#
# --- UVM Report Summary ---
#
# ** Report counts by severity
# UVM_INFO :    9
# UVM_WARNING :    0
# UVM_ERROR :    0
# UVM_FATAL :    0
# ** Report counts by id
# [Questa UVM]     2
# [RNTST]     1
# [TEST_DONE]     1
# [pdm_saturation_seq]     2
# [pdm_saturation_test]     3
# ** Note: $finish    :
/CMC/tools/siemens/questasim_2022.1_2/questasim/linux_x86_64/../verilog_src/uvm-1.1d/src/base/uv
m_root.svh(430)
#    Time: 11040 ns  Iteration: 54  Instance: /top
# End time: 15:03:43 on Nov 08,2026, Elapsed time: 0:00:01
# Errors: 0, Warnings: 0
```

## Running pdm_reset_test.sv terminal output

None

```
Directory 'work' already exists. Skipping vlib.
QuestaSim-64 vmap 2022.1_2 Lib Mapping Utility 2022.04 Apr  2 2022
vmap work work
Modifying
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/modelsim.ini
Compiling design and verification files...
QuestaSim-64 vlog 2022.1_2 Compiler 2022.04 Apr  2 2022
Start time: 15:09:32 on Nov 08,2026
vlog -sv -L mtiUvm
"+incdir+/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verifica
tion"
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/dut/pdm_core.svp
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/dut/pdm_top.v
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
upstream_if.sv
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
downstream_if.sv
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
pkg.sv
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/top.
sv
-- Compiling module pdm
-- Compiling interface pdm_upstream_if
-- Compiling interface pdm_downstream_if
-- Compiling package pdm_pkg
-- Importing package mtiUvm.uvm_pkg (uvm-1.1d Built-in)
** Note: (vlog-2286)
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
pkg.sv(8): Using implicit
+incdir+/CMC/tools/siemens/questasim_2022.1_2/questasim/uvm-1.1d/../verilog_src/uvm-1.1d/src
from import uvm_pkg
-- Compiling module top
-- Importing package pdm_pkg

Top level modules:
        top
End time: 15:09:33 on Nov 08,2026, Elapsed time: 0:00:01
Errors: 0, Warnings: 0
Starting Questa Simulator in BATCH mode for test: pdm_reset_test...
Reading pref.tcl

# 2022.1_2

# set TESTNAME pdm_reset_test
# pdm_reset_test
#  do
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/script/run.do
# ============================================================
```

```
#   RUN.DO: Launching UVM test: pdm_reset_test
# ============================================================
# vsim -L mtiUvm work.top "+UVM_TESTNAME=pdm_reset_test" "+UVM_NO_RELNOTES"
# Start time: 15:09:34 on Nov 08,2026
# ** Note: (vsim-8009) Loading existing optimized design _opt2
# //  Questa Sim-64
# //  Version 2022.1_2 linux_x86_64 Apr  2 2022
# //
# //  Copyright 1991-2022 Mentor Graphics Corporation
# //  All Rights Reserved.
# //
# //  QuestaSim and its associated documentation contain trade
# //  secrets and commercial or financial information that are the property of
# //  Mentor Graphics Corporation and are privileged, confidential,
# //  and exempt from disclosure under the Freedom of Information Act,
# //  5 U.S.C. Section 552. Furthermore, this information
# //  is prohibited from disclosure under the Trade Secrets Act,
# //  18 U.S.C. Section 1905.
# //
# Loading sv_std.std
# Loading work.pdm_upstream_if(fast)
# Loading work.pdm_downstream_if(fast)
# Loading mtiUvm.uvm_pkg(fast)
# Loading work.pdm_pkg(fast)
# Loading mtiUvm.questa_uvm_pkg(fast)
# Loading work.top(fast)
# Loading work.pdm_downstream_if(fast__2)
# Loading /CMC/tools/siemens/questasim_2022.1_2/questasim/uvm-1.1d/linux_x86_64/uvm_dpi.so
# ----------------------------------------------------------------
# UVM-1.1d
# (C) 2007-2013 Mentor Graphics Corporation
# (C) 2007-2013 Cadence Design Systems, Inc.
# (C) 2006-2013 Synopsys, Inc.
# (C) 2011-2013 Cypress Semiconductor Corp.
# ----------------------------------------------------------------
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(277) @ 0: reporter [Questa UVM]
QUESTA_UVM-1.2.3
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(278) @ 0: reporter [Questa UVM]
questa_uvm::init(+struct)
# UVM_INFO @ 0: reporter [RNTST] Running test pdm_reset_test...
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
reset_test.sv(25) @ 110000: uvm_test_top [pdm_reset_test] --- Executing asynchronous reset
sequence (between & during packets) ---
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
reset_traffic_seq.sv(20) @ 110000: uvm_test_top.env.upstream_agent.sequencer@@rst_seq
[pdm_reset_traffic_seq] Starting asynchronous reset test sequence
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
```

```
reset_traffic_seq.sv(26) @ 110000: uvm_test_top.env.upstream_agent.sequencer@@rst_seq
[pdm_reset_traffic_seq] [Iter 0] Step 1: Sending pre-reset packet
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
reset_traffic_seq.sv(33) @ 420000: uvm_test_top.env.upstream_agent.sequencer@@rst_seq
[pdm_reset_traffic_seq] [Iter 0] Step 2: Injecting standalone reset between packets
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
upstream_monitor.sv(53) @ 460000: uvm_test_top.env.upstream_agent.monitor [PDM_RST] Asynchronous
reset observed
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
reset_traffic_seq.sv(42) @ 500000: uvm_test_top.env.upstream_agent.sequencer@@rst_seq
[pdm_reset_traffic_seq] [Iter 0] Step 3: Verifying recovery post-standalone reset
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
reset_traffic_seq.sv(52) @ 660000: uvm_test_top.env.upstream_agent.sequencer@@rst_seq
[pdm_reset_traffic_seq] [Iter 0] Step 4: Injecting mid-transfer reset
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
upstream_monitor.sv(53) @ 760000: uvm_test_top.env.upstream_agent.monitor [PDM_RST] Asynchronous
reset observed
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
reset_traffic_seq.sv(63) @ 820000: uvm_test_top.env.upstream_agent.sequencer@@rst_seq
[pdm_reset_traffic_seq] [Iter 0] Step 5: Verifying clean recovery post-abort
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
reset_traffic_seq.sv(26) @ 980000: uvm_test_top.env.upstream_agent.sequencer@@rst_seq
[pdm_reset_traffic_seq] [Iter 1] Step 1: Sending pre-reset packet
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
reset_traffic_seq.sv(33) @ 1140000: uvm_test_top.env.upstream_agent.sequencer@@rst_seq
[pdm_reset_traffic_seq] [Iter 1] Step 2: Injecting standalone reset between packets
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
upstream_monitor.sv(53) @ 1180000: uvm_test_top.env.upstream_agent.monitor [PDM_RST]
Asynchronous reset observed
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
reset_traffic_seq.sv(42) @ 1260000: uvm_test_top.env.upstream_agent.sequencer@@rst_seq
[pdm_reset_traffic_seq] [Iter 1] Step 3: Verifying recovery post-standalone reset
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
reset_traffic_seq.sv(52) @ 1580000: uvm_test_top.env.upstream_agent.sequencer@@rst_seq
[pdm_reset_traffic_seq] [Iter 1] Step 4: Injecting mid-transfer reset
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
upstream_monitor.sv(53) @ 1840000: uvm_test_top.env.upstream_agent.monitor [PDM_RST]
Asynchronous reset observed
```

```
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
reset_traffic_seq.sv(63) @ 1880000: uvm_test_top.env.upstream_agent.sequencer@@rst_seq
[pdm_reset_traffic_seq] [Iter 1] Step 5: Verifying clean recovery post-abort
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
reset_traffic_seq.sv(26) @ 2040000: uvm_test_top.env.upstream_agent.sequencer@@rst_seq
[pdm_reset_traffic_seq] [Iter 2] Step 1: Sending pre-reset packet
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
reset_traffic_seq.sv(33) @ 2360000: uvm_test_top.env.upstream_agent.sequencer@@rst_seq
[pdm_reset_traffic_seq] [Iter 2] Step 2: Injecting standalone reset between packets
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
upstream_monitor.sv(53) @ 2400000: uvm_test_top.env.upstream_agent.monitor [PDM_RST]
Asynchronous reset observed
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
reset_traffic_seq.sv(42) @ 2460000: uvm_test_top.env.upstream_agent.sequencer@@rst_seq
[pdm_reset_traffic_seq] [Iter 2] Step 3: Verifying recovery post-standalone reset
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
reset_traffic_seq.sv(52) @ 2780000: uvm_test_top.env.upstream_agent.sequencer@@rst_seq
[pdm_reset_traffic_seq] [Iter 2] Step 4: Injecting mid-transfer reset
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
upstream_monitor.sv(53) @ 3040000: uvm_test_top.env.upstream_agent.monitor [PDM_RST]
Asynchronous reset observed
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
reset_traffic_seq.sv(63) @ 3100000: uvm_test_top.env.upstream_agent.sequencer@@rst_seq
[pdm_reset_traffic_seq] [Iter 2] Step 5: Verifying clean recovery post-abort
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
reset_traffic_seq.sv(26) @ 3260000: uvm_test_top.env.upstream_agent.sequencer@@rst_seq
[pdm_reset_traffic_seq] [Iter 3] Step 1: Sending pre-reset packet
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
reset_traffic_seq.sv(33) @ 3580000: uvm_test_top.env.upstream_agent.sequencer@@rst_seq
[pdm_reset_traffic_seq] [Iter 3] Step 2: Injecting standalone reset between packets
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
upstream_monitor.sv(53) @ 3620000: uvm_test_top.env.upstream_agent.monitor [PDM_RST]
Asynchronous reset observed
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
reset_traffic_seq.sv(42) @ 3700000: uvm_test_top.env.upstream_agent.sequencer@@rst_seq
[pdm_reset_traffic_seq] [Iter 3] Step 3: Verifying recovery post-standalone reset
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
```

```
reset_traffic_seq.sv(52) @ 3940000: uvm_test_top.env.upstream_agent.sequencer@@rst_seq
[pdm_reset_traffic_seq] [Iter 3] Step 4: Injecting mid-transfer reset
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
upstream_monitor.sv(53) @ 4220000: uvm_test_top.env.upstream_agent.monitor [PDM_RST]
Asynchronous reset observed
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
reset_traffic_seq.sv(63) @ 4280000: uvm_test_top.env.upstream_agent.sequencer@@rst_seq
[pdm_reset_traffic_seq] [Iter 3] Step 5: Verifying clean recovery post-abort
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
reset_traffic_seq.sv(73) @ 4440000: uvm_test_top.env.upstream_agent.sequencer@@rst_seq
[pdm_reset_traffic_seq] Completed asynchronous reset sequence
# UVM_INFO verilog_src/uvm-1.1d/src/base/uvm_objection.svh(1267) @ 5040000: reporter [TEST_DONE]
'run' phase is ready to proceed to the 'extract' phase
#
# ============================================================
#                   TEST EXECUTION SUMMARY
# ============================================================
#   TEST NAME    : pdm_reset_test
#   SIM TIME     : 5040000
#   FATAL COUNT  : 0
#   ERROR COUNT  : 0
#   WARNING COUNT: 0
#   INFO COUNT   : 35
# ------------------------------------------------------------
#               *** UVM TEST PASSED ***
# ============================================================
#
#
# --- UVM Report Summary ---
#
# ** Report counts by severity
# UVM_INFO :   35
# UVM_WARNING :    0
# UVM_ERROR :    0
# UVM_FATAL :    0
# ** Report counts by id
# [PDM_RST]     8
# [Questa UVM]     2
# [RNTST]     1
# [TEST_DONE]     1
# [pdm_reset_test]     1
# [pdm_reset_traffic_seq]    22
# ** Note: $finish    :
/CMC/tools/siemens/questasim_2022.1_2/questasim/linux_x86_64/../verilog_src/uvm-1.1d/src/base/uv
m_root.svh(430)
#    Time: 5040 ns  Iteration: 54  Instance: /top
# End time: 15:09:35 on Nov 08,2026, Elapsed time: 0:00:01
```

```
# Errors: 0, Warnings: 0
```

## Running pdm_random_test.sv terminal output

```
None
Directory 'work' already exists. Skipping vlib.
QuestaSim-64 vmap 2022.1_2 Lib Mapping Utility 2022.04 Apr  2 2022
vmap work work
Modifying
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/modelsim.ini
Compiling design and verification files...
QuestaSim-64 vlog 2022.1_2 Compiler 2022.04 Apr  2 2022
Start time: 15:10:07 on Nov 08,2026
vlog -sv -L mtiUvm
"+incdir+/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verifica
tion"
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/dut/pdm_core.svp
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/dut/pdm_top.v
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
upstream_if.sv
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
downstream_if.sv
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
pkg.sv
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/top.
sv
-- Compiling module pdm
-- Compiling interface pdm_upstream_if
-- Compiling interface pdm_downstream_if
-- Compiling package pdm_pkg
-- Importing package mtiUvm.uvm_pkg (uvm-1.1d Built-in)
** Note: (vlog-2286)
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
pkg.sv(8): Using implicit
+incdir+/CMC/tools/siemens/questasim_2022.1_2/questasim/uvm-1.1d/../verilog_src/uvm-1.1d/src
from import uvm_pkg
-- Compiling module top
-- Importing package pdm_pkg

Top level modules:
        top
End time: 15:10:07 on Nov 08,2026, Elapsed time: 0:00:00
Errors: 0, Warnings: 0
Starting Questa Simulator in BATCH mode for test: pdm_random_test...
Reading pref.tcl

# 2022.1_2
```

```
# set TESTNAME pdm_random_test
# pdm_random_test
#  do
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/script/run.do
# ============================================================
#   RUN.DO: Launching UVM test: pdm_random_test
# ============================================================
# vsim -L mtiUvm work.top "+UVM_TESTNAME=pdm_random_test" "+UVM_NO_RELNOTES"
# Start time: 15:10:08 on Nov 08,2026
# ** Note: (vsim-8009) Loading existing optimized design _opt2
# //  Questa Sim-64
# //  Version 2022.1_2 linux_x86_64 Apr  2 2022
# //
# //  Copyright 1991-2022 Mentor Graphics Corporation
# //  All Rights Reserved.
# //
# //  QuestaSim and its associated documentation contain trade
# //  secrets and commercial or financial information that are the property of
# //  Mentor Graphics Corporation and are privileged, confidential,
# //  and exempt from disclosure under the Freedom of Information Act,
# //  5 U.S.C. Section 552. Furthermore, this information
# //  is prohibited from disclosure under the Trade Secrets Act,
# //  18 U.S.C. Section 1905.
# //
# Loading sv_std.std
# Loading work.pdm_upstream_if(fast)
# Loading work.pdm_downstream_if(fast)
# Loading mtiUvm.uvm_pkg(fast)
# Loading work.pdm_pkg(fast)
# Loading mtiUvm.questa_uvm_pkg(fast)
# Loading work.top(fast)
# Loading work.pdm_downstream_if(fast__2)
# Loading /CMC/tools/siemens/questasim_2022.1_2/questasim/uvm-1.1d/linux_x86_64/uvm_dpi.so
# ----------------------------------------------------------------
# UVM-1.1d
# (C) 2007-2013 Mentor Graphics Corporation
# (C) 2007-2013 Cadence Design Systems, Inc.
# (C) 2006-2013 Synopsys, Inc.
# (C) 2011-2013 Cypress Semiconductor Corp.
# ----------------------------------------------------------------
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(277) @ 0: reporter [Questa UVM]
QUESTA_UVM-1.2.3
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(278) @ 0: reporter [Questa UVM]
questa_uvm::init(+struct)
# UVM_INFO @ 0: reporter [RNTST] Running test pdm_random_test...
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
random_test.sv(31) @ 110000: uvm_test_top [pdm_random_test] --- Running full constrained-random
test (30 packets) ---
```

```
# UVM_INFO
/home/.engr-ece/rfp479/Documents/02-CME-435/CME435_LAB4/02_CME435_L4/CME435_L4/verification/pdm_
random_traffic_seq.sv(20) @ 110000: uvm_test_top.env.upstream_agent.sequencer@@seq
[pdm_random_traffic_seq] Generating 30 constrained-random packets
# UVM_INFO verilog_src/uvm-1.1d/src/base/uvm_objection.svh(1267) @ 9380000: reporter [TEST_DONE]
'run' phase is ready to proceed to the 'extract' phase
#
# ------------------------------------------------------
#                  TEST EXECUTION SUMMARY
# ------------------------------------------------------
#   TEST NAME    : pdm_random_test
#   SIM TIME     : 9380000
#   FATAL COUNT  : 0
#   ERROR COUNT  : 0
#   WARNING COUNT: 0
#   INFO COUNT   : 6
# ------------------------------------------------------
#               *** UVM TEST PASSED ***
# ------------------------------------------------------
#
#
# --- UVM Report Summary ---
#
# ** Report counts by severity
# UVM_INFO :    6
# UVM_WARNING :    0
# UVM_ERROR :    0
# UVM_FATAL :    0
# ** Report counts by id
# [Questa UVM]     2
# [RNTST]     1
# [TEST_DONE]     1
# [pdm_random_test]     1
# [pdm_random_traffic_seq]     1
# ** Note: $finish    :
/CMC/tools/siemens/questasim_2022.1_2/questasim/linux_x86_64/../verilog_src/uvm-1.1d/src/base/uv
m_root.svh(430)
#    Time: 9380 ns  Iteration: 54  Instance: /top
# End time: 15:10:10 on Nov 08,2026, Elapsed time: 0:00:02
# Errors: 0, Warnings: 0
```

Submission Instructions

- Zip your project folder (cme435_lab_4) and include any useful information to help the marker.
- Hand in the zip file to Canvas before the specified due date.